# Juhwan Lee

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## SUMMARY

Ph.D. student and research assistant in Air Transportation at Korea Aerospace University, with experience in airport operations, air traffic management, aviation simulation, capacity and delay analysis, and data-driven forecasting. My research focuses on understanding dynamic aircraft and passenger flows and integrating simulation and AI-based prediction into decision support systems to enable proactive, adaptive, and efficient airport operations.

## EDUCATION

**Korea Aerospace University (KAU), Goyang, South Korea** *Sep 2023 – Present*
Ph.D. Student in Air Transportation, Transportation and Logistics
*Research focus: airport operations, capacity analysis, airport simulation, and data-driven forecasting*

**Korea Aerospace University (KAU), Goyang, South Korea** *August 2023*
M.S. in Air Transportation, Transportation and Logistics
GPA: 4.0/4.0 (4.5/4.5)
*Key courses: Introductory Applied Machine Learning, Airport Design with Simulation, Advanced Discrete Choice Theory in Transportation, Theory and Application of Data Mining, Airlines Operations Control*

**Korea Aerospace University (KAU), Goyang, South Korea** *August 2021*
B.S. in Air Transportation, Transportation and Logistics
GPA: 3.77/4.0 (4.05/4.5)
*Key courses: Air Traffic Management, Airspace System Analysis, Air Transportation System Analysis, Aviation Weather Service, Flight Operation Planning & Control, Computer Applications for Air Transportation Studies, Airport Operations*

## RESEARCH INTERESTS

**Data-Driven Airport Operations and Aviation Decision Support**

- Airport capacity and demand–capacity imbalance analysis
- Passenger terminal simulation and data-driven congestion forecasting
- AI-enabled decision support for airport operations and resource allocation

## PUBLICATIONS

[1] **Juhwan Lee**, S. Yoon, K. Lee, S. Jeong, and H. Baik, "Airport Passenger Queue Forecasting for Departure Gates and Security Checkpoints," IEEE/AIAA 45th Digital Avionics Systems Conference (DASC), 2026.

[2] J. Shin, H. Baik, **Juhwan Lee**, and B. Yi, "Network Connectivity Analysis of Major Asian Airports Using the NetScan Model:Focusing on Incheon Internation Airport," Journal of the Korean Society for Aviation and Aeronautics, 2025.

[3] J. Shin, B. Yi, M. Kim, J. Cho, and **Juhwan Lee**, "Study on Airport Capacity Utilization Analysis Using the Capacity Coverage Chart (CCC)," Journal of the Korean Society for Aviation and Aeronautics, 2025.

[4] E. Kim, **Juhwan Lee**, Y. Shin, and H. Baik, "A Study on Airport Access Mode Choice Behavior Using Mobile Data: Case of Incheon International Airport," IEEE/AIAA 42nd Digital Avionics Systems Conference (DASC), 2023.

[5] Y. Lee, **Juhwan Lee**, and K. Lee, "A Study on the Visualization of an Airline's Fleet State Variation," *Journal of the Korean Society for Aviation and Aeronautics*, 2021.

## PROJECT EXPERIENCE

**Integrated Information System Enhancement Project**
(Sep 2025 – Mar 2026)

- Analyzed airport operational data, including flight schedules, passenger flows, and observed queue lengths and waiting times.
- Developed and validated a CAST-based passenger terminal simulation model.

**Strategic Planning for Enhancing Hub Competitiveness of Incheon International Airport**
(Oct 2025 – Dec 2025) [2],[3]

- Analyzed changes in delay according to passenger demand and capacity conditions.
- Supported strategic evaluation of airport competitiveness from an operational performance perspective.

**Development of Passenger Terminal Simulation Models for Incheon International Airport**
(Jul 2025 – Dec 2025)

- Developed MATLAB-based simulations for departure processing facilities using actual terminal data.
- Estimated queue length and waiting time under time-varying demand and operational conditions.

**Analysis of Variable Multiple Taxi-Line Operations for Apron Congestion Mitigation at Jeju International Airport**
(Jul 2025 – Dec 2025)

- Simulated apron congestion and aircraft surface movement using TAAM.
- Evaluated taxi-line operational alternatives for congestion mitigation and surface movement efficiency.

**Master Plan for Incheon International Airport**
(Sep 2022 – Aug 2024) [4]

- Performed runway capacity analysis using TAAM and passenger terminal simulation using CAST.
- Evaluated capacity, congestion, delay, and economic feasibility for airport master planning.

**Performance-Based Navigation (PBN) Flight Procedure Maintenance Service**
(Sep 2021 – Dec 2021)

- Analyzed observed aircraft trajectory data to evaluate the effectiveness of flight procedures under the operation of Continuous Descent Operations (CDO) and Continuous Climb Operations (CCO).

## AWARDS AND ACHIEVEMENTS

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| Scholarship, KAU Professional Talent Development Scholarship | $4,520 | *Fall 2021* |
| 2nd Place in Capstone Design Competition | $260 | *Fall 2020* |
| 2nd Place in ICT Startup Makerthon | $740 | *Fall 2019* |
| Scholarship, Jeongseok C | $841 | *Fall 2019* |
| Scholarship, Jeongseok C | $739 | *Fall 2016* |

## TECHNICAL SKILLS

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|---|---|
| **Programming** | Python, MATLAB, LaTeX |
| **AI / Data Analysis** | PyTorch, Time-series forecasting, Transformer-based modeling, Statistical analysis |
| **Simulation** | TAAM (Total Airspace and Airport Modeller), CAST (Passenger terminal simulation) |
| **Aviation Analysis** | Airport capacity and delay analysis, Queue and waiting time analysis |